Problem A – Let's count words

Two words W_1 and W_2 are said to be generated from the same word W, if the three words have the same length L, and you can read both W_1 and W_2 from W reading exactly L letters starting at a position in W. If while reading from W you get to the end of the word you start again from the first position. For example *rdwo* and *ordw* are both generated from the same word *word*, the first one is if you start reading from the third position of the word, the second one is if you start from the second position. A word W is generated from the same word W as you can read the word starting at the first position of it.

Given a list of N words, can you find how many different generator words W were used to create the list?

Input

The first line of input contains a single integer N ($1 \leq N \leq 100$), representing the number of words in the list. The second and last line of input contains N strings separated by a space, representing the N words in the list. Each word in the list will contain only lower case letters from the english alphabet and will have no more than 100 letters.

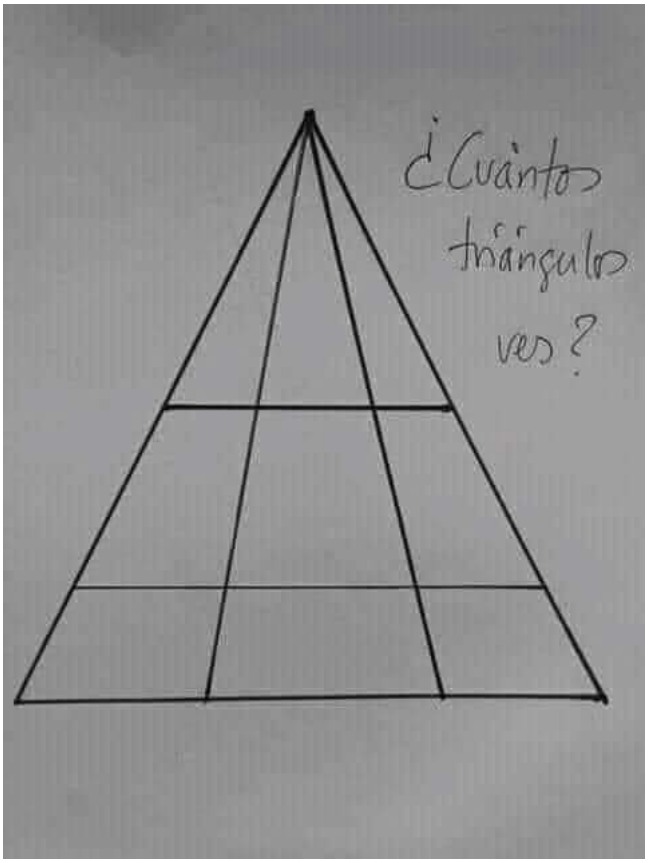
Output

Print a single line with an integer, representing the number of different generator words W that were used to create the list.

Sample input 1	**Sample output 1**
`7` `word icpc rdwo dwor ordw pcic ccpi`	`3`

Problem B – Counting triangles

Can you find how many triangles are in the following drawing?



Your task is to count the number of triangles in a drawing similar to the one above, where the outer triangle has N lines inside it going from the top to the base, and K lines inside that are parallel to its base.

Input

The first line contains an integer T $1 \leq T \leq 10^5$, representing the number of test cases. Each of the next T lines represents a test case containing two integers separated by a space, N and K ($0 \leq N \leq 10^9$, $0 \leq K \leq 10^9$), representing, respectively, the number of lines going from the top to the base and the number of lines parallel to the base in the drawing.

Output

For each test case print the number of triangles in the drawing. As this number can be big print it mod 1000000007

Sample input 1	**Sample output 1**
2 2 2 3 1	18 20

Problem C – Debugging the network

A communication string is defined as a string containing only lowercase letters from the English alphabet (‘a’, ‘b’, ‘c’, ‘d’, ‘e’, ‘f’, ‘g’, ‘h’, ‘i’, ‘j’, ‘k’, ‘l’, ‘m’, ‘n’, ‘o’, ‘p’, ‘q’, ‘r’, ‘s’, ‘t’, ‘u’, ‘v’, ‘w’, ‘x’, ‘y’, ‘z’). In order to make easier and cheaper to send these strings over slow networks a compression method has been defined as follow: If a substring in the string with length of at least 2 characters consists only of occurrences of the same letter, then, replace the substring with its length followed by the letter, this is done always in a way that no two consecutive positions of the string have the same letter. For example, the string *aaaaabbc* would be replaced with the string *5a2bc*. The communication string is compressed before sending the message over the network, then a system receiving the message will decompress the string before the communication is acknowledged to be successful.

After years of this compression method working as expected suddenly some systems in the network started to break, some debugging in the system showed it breaks when the communication string is longer than K characters.

Your task is to determine if the system in the network will not break while uncompressing a compressed string given the compressed string, and the value K.

Input

The first line of input contains a single integer T ($1 \leq T \leq 100$), representing the number of test cases. Each of the next T lines contain a string S ($1 \leq |S| \leq 1000$) and a number K ($1 \leq K \leq 10^6$) separated by a space, representing, respectively, the compressed string and the longest communication string length that is able to be transmitted over the network.

Output

For each test case in the input print a line containing the communication string if the given compressed string is safe to be uncompressed, otherwise print a line with the string “unfeasible”.

Sample input 1	**Sample output 1**
4 5a2bc 8 5a2bc 7 asdf4x 50 asjkdf10000000000kz 1000000	aaaaabbc unfeasible asdfxxxx unfeasible

Kitchen Waste

Input file:	`standard input`
Output file:	`standard output`
Time limit:	1 second
Memory limit:	256 megabytes

As some of you may know, Jaime, the delivery package company owner, is also owner of a famous restaurant at Quadradonia. Jaime's restaurant iconic plate is an exquisite clam chowder, which is served inside a sourdough bread that is used as a bowl. The clam chowder and the sourdough bread are homemade using zero waste cooking, which means literally, to have no waste left behind while cooking a meal. What is not zero waste is the way the kitchen staff serves the bowls.

For tonights service the restaurant staff expects to serve N sourdough breads of the clam chowder pouring it from the M dutch ovens where the clam chowder was prepared. To make their work easier, the kitchen staff have placed the N breads in a line, in such way they always know what is the next bread to be filled with soup, since all bread is homemade, not all of them may hold the same amount of soup. Also, there will be only one person serving the soup from a dutch oven at the time, so all the content that can be used from a dutch oven is used before taking the next dutch oven to use for the soup service, similar to what was done with the bread, they placed the M dutch ovens in a line, so that when the person serving the soup decides to change the dutch oven, they only have to take the next dutch oven in line.

When the service starts, the first dutch oven from the line of dutch ovens is prepared for serving, and the first clam chowder should be served into the first bread from the line of breads. Before serving the soup, the staff will check, if the dutch oven has enough soup to fill the bread, then, they serve it, otherwise they will take the next dutch oven of soup from the line of dutch ovens and send the current dutch oven to the dishwasher, throwing the clam chowder left on the dutch oven as discard. Once all the N breads are served, the service is over, and, as clam chowder should be fresh every night, all the soup in the current dutch oven is discarded, and the same is done to all dutch ovens from the line that were not used. As you can see, this way of serving the chowder may tend to have a lot of waste.

Jaime wants to measure how wasteful this method is, so he came to you and hired you to measure the waste. Given the volume of soup each dutch oven in the line of dutch ovens holds, and the volume of soup each sourdough bread in the line of breads needs to be filled, find how much soup will be wasted during the soup serving service.

Input

The first line of input contains two integers separated by a space N and M ($1 \leq N \leq M \leq 1000$), representing the number of sourdough breads to serve, and the number of dutch ovens in the line. The next line contains N integer numbers separated by a space, where the i-th number represent the volume needed to fill the i-th bread in the line, this volume will be a number between 1 and 20. The next and last line contains M integer numbers separated by a space, where the i-th number represents the volume of soup the i-th dutch oven has, this volume will be a number between 20 and 100. It is guaranteed all breads can be served with the given input.

Output

Output a line containing one integer number, the amount of waste Jaime's kitchen staff will waste in the service.

Examples

standard input	standard output
5 5 20 20 20 20 20 20 20 20 20 20	0
5 6 1 2 3 4 5 20 20 20 20 20 20	105

Lets Count Factors

Input file:	`standard input`
Output file:	`standard output`
Time limit:	1 second
Memory limit:	256 megabytes

Planet E-13 orbits around a star in a faraway galaxy named UAZ. All University students in that planet need to know well how to perform prime factorization on numbers, otherwise they risk failing the courses they take. One of the teachers likes asking the students in his class the following question: how many different prime numbers must be multiplied together (maybe some of them more than once) to obtain the same result as the product of a pair of the given numbers A and B?

Thanitos is a student taking such course and he is really nervous; he does not want to fail. Your task is to help Thanitos answer N teachers question.

Input

The first line contains the number N ($1 \leq N \leq 10^5$) of questions that the teacher will make. Each of the following N lines will contain a question the teacher makes and it consists of the numbers A and B ($1 \leq A, B \leq 10^7$) separated by a space.

Output

For each teacher question print a line with a number representing how many different prime numbers must be multiplied to obtain A*B (the product of A and B).

Example

standard input	standard output
5 13 17 120 980 43 1 60 1 78 11	2 4 1 3 4

Goombas Colliding

Input file:	`standard input`
Output file:	`standard output`
Time limit:	0.5 seconds
Memory limit:	256 megabytes

Atsa just bought Super Mario Maker and wants to test your skills for an analysis with a level that he prepared.

That level consists of a platform with some Goombas in it. As you know, Goombas are characters with the following behavior:

- Initially, they point in one direction, (left: 0, right: 1).
- They move in the direction they are currently facing, as long as there are no obstacles.
- If two Goombas collide, they will flip their direction inmediately.
- When a Goomba reaches one end of the platform, it falls.

The platform is L blocks long, extending from the coordinate 0, to L. Above it, there will be G Goombas. The $i-th$ Goomba will be located at p_i facing to the direction d_i. All Goombas will advance with a speed of 1 block per second.

Atsa wants you to tell him how many seconds it will take for the platform to be empty.

For this problem purposes, consider the size of a Goomba to be a single point. No two Goombas will share the same initial x coordinate.

Input

The first line of the input contains two integers L $\left(2 \leq L \leq 10^{16}\right)$ and G $\left(1 \leq G \leq 10^{4}\right)$.

Each of the following G lines contains two integers p_i $(0 < p_i < L)$ and d_i $(d_i \in \{0, 1\})$.

Output

The time in seconds that must pass for the platform to be empty.

Examples

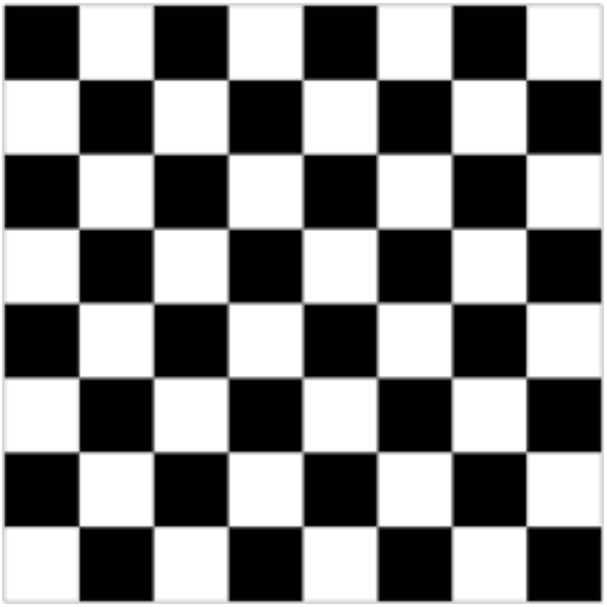
standard input	standard output
3 2 1 1 2 0	2
5 2 1 0 2 1	3

Atsa's Checkers Board

Input file:	`standard input`
Output file:	`standard output`
Time limit:	0.5 seconds
Memory limit:	256 megabytes

Atsa is constantly looking for new ways to have fun during this quarantine, she used to play checkers against her friends but, after winning against each of them, no one wants to play checkers against her anymore. Atsa now considers to play checkers alone, but found a more interesting way to challenge herself.

A checkers board is represented by a rectangular grid with N rows and M columns, with a total of NxM cells. Each cell has a light color, or a dark color, in such way that no adjacent cells have the same color, the figure below shows a checkers board with 8 rows and 8 columns:



Since Atsa has a lot of checkers boards, she also has a lot of stones to play checkers, she has enough stones to place at least one stone on each cell of any of her checkers boards. Similar to the cells in the board, each stone is either light colored or dark colored. Atsa found that if she places a light colored stone on each light colored cell of the board, and a dark colored stone on each dark colored cell, then all squares of 2x2 cells in the board will have two dark colored stones, and two light colored stones, she defined this configuration as a "pairable" stones configurations. Atsa believes this is not the only "pairable" stones configuration in the board and decided to find all of them on each of the boards she has. Are you be able to find how many distinct "pairable" stones configurations can be placed in a checkers board?

Input

The first and only line of input contains two integer numbers separated by a space, representing respectively, the number N of rows in the board, and the number M of columns in the board ($1 \leq N, M \leq 60$).

Output

Output a single line with a single number, representing, the number of distinct "pairable" stones configurations that Atsa can place in the checkers board given in the input.

Examples

standard input	standard output
2 2	6
2 3	10

Problem H – Leonel and the powers of two

Leonel is solving exponentiation problems, he has become really good to calculate powers of two. Several times Leonel has impressed his friends and teacher showing his skills answering what is the value for the k-th power of two (2^k), with really large values for k.

To make things more challenging to Leonel, his teacher has requested to him, instead of calculating the value for 2^k, to write the result in the following notation:

- 2 if $k = 1$
- $(2 * 2^{k-1})$ if k is an odd number
- $(2^{\frac{k}{2}})^2$ if k is an even number

Leonel will have to repeat the process until no more changes can be done to the current notation. For example if $k = 5$, then the first time Leonel does the notation he will write $(2 * 2^4)$. Leonel can change the notation replacing 2^4 with $(2^{\frac{4}{2}})^2 = (2^2)^2$, resulting in : $(2 * (2^2)^2)$, Leonel can change this notation replacing 2^2 with $(2^{\frac{2}{2}})^2$, after this change the notation will be: $(2*((2^1)^2)^2)$, one last change can be done to the notation changing 2^1 with 2, which will result in $(2*((2)^2)^2)$, since no more changes can be done, Leonel has found the answer for $k = 5$.

Leonel is certain it will take more time for him to write in this notation than calculating the power, that is why he decided to ask for your help to write a computer program so he can copy the notation from the result of your program to his notebook. Given the value of k write a program that prints 2^k in the notation described by Leonel's teacher.

Input

The first line of input contains a single integer T ($1 \leq T \leq 10^4$), the number of test cases. Each of the next T lines contains a single integer k ($1 \leq k \leq 10^{18}$), representing the power of two Leonel has to write in his teacher's notation.

Output

For each test case in the input print the result of the final notation Leonel will get.

Sample input 1	**Sample output 1**
5 1 2 3 4 5	2 (2)^2 (2*(2)^2) ((2)^2)^2 (2*((2)^2)^2)

Problem I – Escape Room

A new escape room has opened in the city. In this escape room, a group of at least two people is required to play. At the beginning of the game one person of the group will be selected, this person will enter the room while others stay outside of the room with a map. The one that enters the room has a device and will be blindfolded, so, the person is not aware of the surroundings. The game is as follows, the person inside the room will use the device to tell the others in the group the coordinates (i, j) the person believes is positioned in the map, the group then will answer with the steps the person should follow in order to escape the room.

The map of the room is represented by a matrix with R rows and C columns, each position of the matrix has a character wich means the following:

- '.' means the person can be positioned in that position.
- '#' means there is a wall at that position.
- 'X' means there is an object at that position.
- 'E' represents the position of the exit in the room.

For each coordinate the person in the room sends to the group, the group should answer with a single character:

- 'W' if the coordinate sent has a wall in the map.
- 'X' if the coordinate sent has an object in the map.
- 'E' if the coordinate sent is the exit.
- '?' If the coordinate indicates a place with no way to the exit
- An indication of the path the person should follow to get to the exit if the map has a '.' in that coordinate.

The indications of the path the group will send to the person are always in such way that if the person moves in that direction the person will be closer to the exit than before, the indication should be a character based on the move the person should perform:

- 'L' if the person should move left.
- 'D' if the person should move down.
- 'R' if the person should move right.
- 'U' if the person should move up.

If the person can move in more than one direction and get closer to the exit, the group should answer with the first from the above list that makes the person closer to the exit. People can't go through object nor walls.

Given the map of the room, and the coordinates sent by the person inside the room, write a program with the answer the group should send to the person.

Input

The first line of input contains two integers separated by a space R and C ($1 \leq R, C \leq 1000$), representing the number of rows and columns in the map. The next R lines in the input contain C characters, the j-th character of the i-th line represents the character in the map at coordinate (i, j). Coordinates start at (1, 1) and end in (R, C). The next line contains a single integer Q, the number of coordinates the person in the room will send. Each of the next Q ($1 \leq Q \leq 10^5$) lines contains two integer numbers separated by a space, representing the coordinate (i, j) sent by the person in the room.

Output

For each coordinate sent by the person print a line with a single character, the answer the group should send to the person.

Sample input 1	**Sample output 1**
5 5 ...## .E..# #.X.# ...## ...#. 5 1 1 2 2 3 3 3 1 5 5	D E X W ?

Problem J – Moon Dancers

The moon dancers, as their name suggests, is a group of people who dances to pray the moon. The moon is a goddess for their tribe, they dance every full moon to get the moon blessings and guarantee the tribe another moon cycle of wellness.

Some researchers are very interested in the peculiar dance these moon dancers perform on the full moon. It is marvelous to see how they dance around a circle, each dancer positioned and performing an exact set of moves that have been passed between dancers for generations. At some point of the dance, each of the N dancers sit at some point in the perimeter of the circle, no two dancers sit in the same place, and they start singing the song of times. Then, suddenly, a set of K dancers stand up and dance rotating counter clockwise around the circle until each of the K dancers meet with a dancer that is sitting, making K pairs of dancers, all of the K dancers will have rotated the same amount of degrees around the circle. At this point the K pairs continue with the dance while the $N - 2 * K$ (possibly none) dancers that are not matched continue singing the song of times.

Knowing your programming skills, researchers have come to you and ask for help on a very specific task: given the position of each of the N dancers when they sit at the perimeter of the circle, can you find the maximum number of pairs that can be matched to continue the dance?

Input

The first line contains a single integer N ($2 \leq N \leq 360$), representing the number of dancers in the tribe. The second and last line contains N integer numbers separated by a space, representing the angle in degrees a_i ($0 \leq a_i < 360$) where the i-th dancer sits, it is guaranteed no two dancers will sit in the same place.

Output

Output a line containing a single integer. The maximum number of pairs that can be matched to continue the dance.

Sample input 1	**Sample output 1**
2 0 1	1

Sample input 2	**Sample output 2**
6 0 1 2 90 91 92	3

Sample input 3	**Sample output 3**
3 359 0 1	1

Problem K – Fixing Subtitles

Jaime is watching a movie, since it's late and all his family is sleeping, he decided to enable subtitles and watch the movie without sound. But, what a surprise, the subtitles are out of sync! He reads the subtitles before some actor in the movie speaks, unacceptable! As the good programmer Jaime is, he decided to look for the file where the subtitles are stored in the movie, he found the file and it's very big! So he will write a program to fix the subtitles and continue watching the movie.

After looking at the file, Jaime found that the subtitles have a well defined structure:

1. A subtitle begins with a line with a number that appears to be a sequence number of the subtitle.
2. The next line contains a string, apparently the time at miliseconds precision where the subtitle should be displayed in the screen and the time when it should be removed from screen separated by the string " `-->` ", the time is displayed in the format "Hours: Minutes:Seconds,Miliseconds".
3. Finally, one or two lines with the text to be displayed as subtitle. Two subtitles are separated each other with a blank line. And the lines with the subtitle text do not have trailing or leading whitespaces.

The subtitles file looks something like this:

```
1
00:00:00,498 --> 00:00:02,827
- Here's what I love most
about food and diet.

2
00:00:02,827 --> 00:00:06,383
We all eat several times a day,

3
00:00:06,383 --> 00:00:09,427
of what goes on our plate
and what stays off.
```

The movie does not last more than 10 hours, and subtitles are at most delayed 10 seconds.

Given the subtitles file and the delay in the subtitles, Can you help Jaime write a program to sync the subtitles of the movie?

Input

The first line of input contains an integer number and a float number with three decimal places, representing the number of subtitles in the file and the delay Jaime found the subtitles have. Then N ($1 \leq N \leq 5000$) subtitles will be given in the input, using the format specified in the problem statement. No subtitle line is longer than 200 characters.

Output

You need to print exactly the same subtitles content, with the adjusted start and end times for each subtitle in the file.

Sample input 1

```
3 3.200
1
00:00:00,498 --> 00:00:02,827
- Here's what I love most
about food and diet.

2
00:00:02,827 --> 00:00:06,383
We all eat several times a day,

3
00:00:06,383 --> 00:00:09,427
of what goes on our plate
and what stays off.
```

Sample output 1

```
1
00:00:03,698 --> 00:00:06,027
- Here's what I love most
about food and diet.

2
00:00:06,027 --> 00:00:09,583
We all eat several times a day,

3
00:00:09,583 --> 00:00:12,627
of what goes on our plate
and what stays off.
```

Problem L – Introducing Teleporting Machine

Nlogonia is connected to Linealonia by a directed linear highway, this is, vehicles can travel from Nlogonia to Linealonia but not in the other way, and the highway does not have any curves, it is a straight line. The highway has N cities, there are no two cities in the same place and, they are numbered from 1 to N in such way that if city j is after city i when you take the highway if $j > i$ for all $i, j \leq N$, city N is Linealonia.

The next year Linealonia will host a programming contest, each city in the highway will send a team to compete in the contest, the cost it takes for a team to go from one city to another is equal to the distance between the two cities. Linealonia has agreed to pay for the travel expenses of all the teams, so they want to find a way to minimize the total cost for all the teams to get to city N. To achieve this, they will build a teleport machine to connect two cities, the only restriction for this machine is that the two cities it connects should be at most at K Km distance away, and if it connects city i and j, city j should be as far as possible from i, considering the K km restriction. The main advantage of building this machine is that if a team decides to use the machine the cost to travel between the two cities the machine connects will be a constant T, hopefully reducing the total cost for Linealonia.

Can you help Linealonia find what two cities should the machine connect in order to reduce the total cost to gather all the teams for their programming contest?

Input

The first line of input contains three integer numbers separated by a space N, K and T, ($1 \leq N \leq 10^5$, $1 \leq K \leq 10^9$, $1 \leq T \leq 10^9$) representing the number of cities in the highway, the maximum distance between the two cities where the teleport machine can be built, and the cost of using the machine. The second and last line of input contains N integer numbers separated by a space, the i-th number represents the km where the i-th city is in the highway. No contiguous cities (i.e. i and $i+1$) are more than 10^9 Km apart and the machine can always be built

Output

If the total cost to gather all the teams at Linealonia can be reduced building the teleport machine print a line with three integer numbers, i, j, and c, representing the cities i and j the machine should connect $j > i$, and the minimum cost c achieved, if there are more than two pairs where the minimum cost can be achieved building the machine, print the one with the minimum i. If the total cost can not be reduced building the teleport machine print -1.

Sample input 1	**Sample output 1**
`4 6 6` `2 6 10 13`	`-1`

Sample input 2	**Sample output 2**
`4 15 1` `2 6 10 13`	`2 4 9`